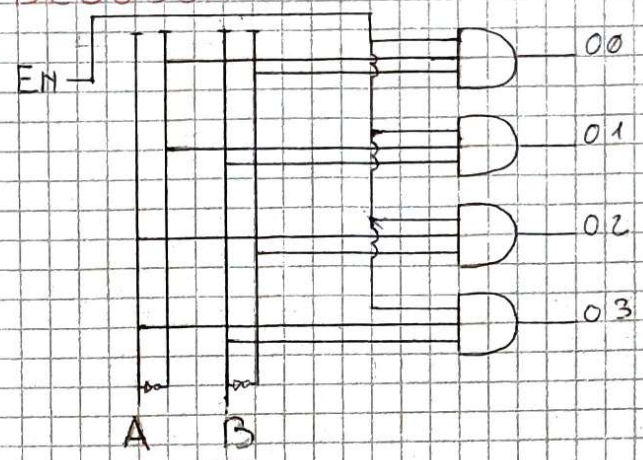
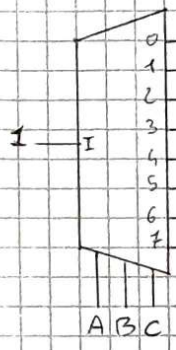
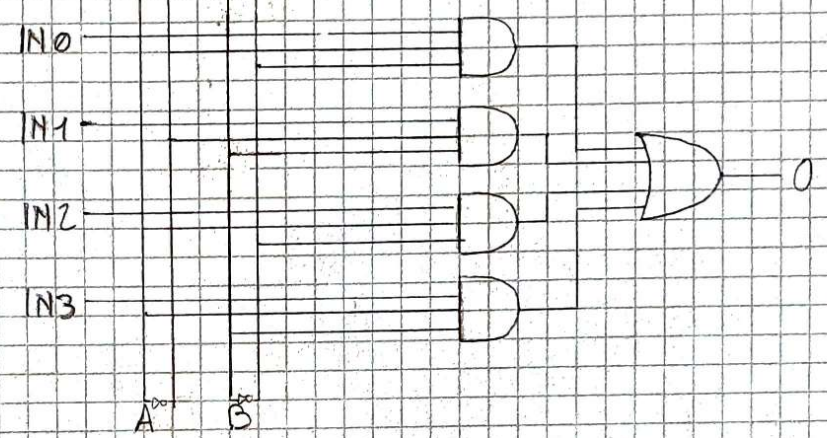
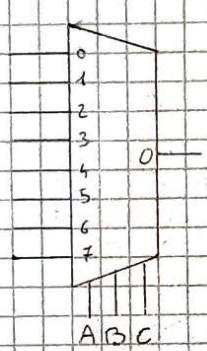


COMPONENTI NOTEVOLI COMBINATORI

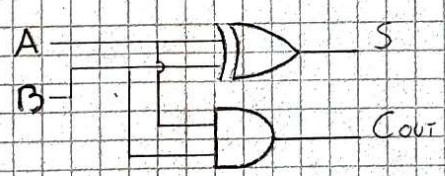
DEMULTIPLEXER / DECODER



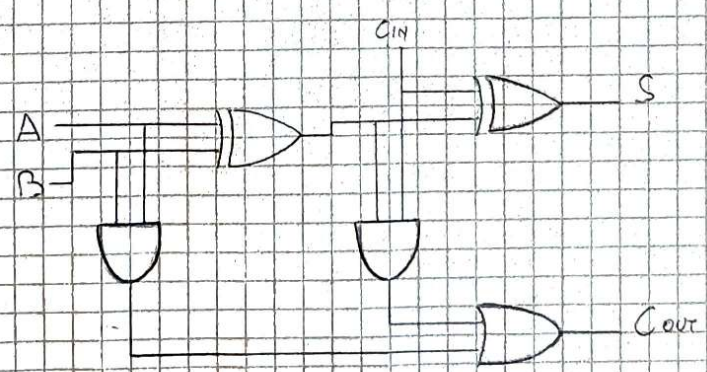
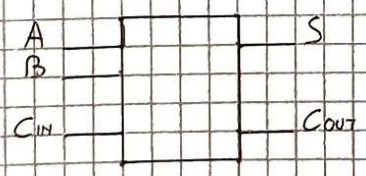
MULTIPLEXER



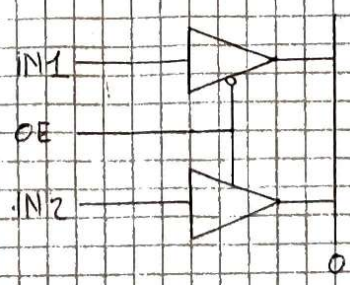
HALF - ADDER



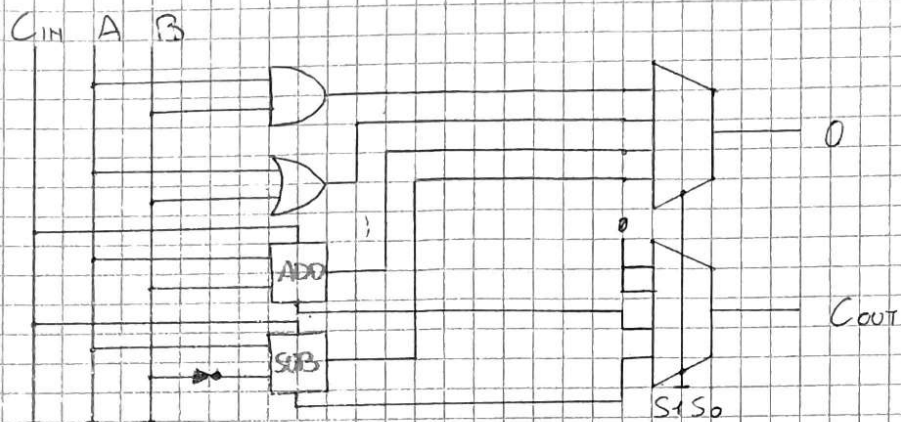
FULL - ADDER



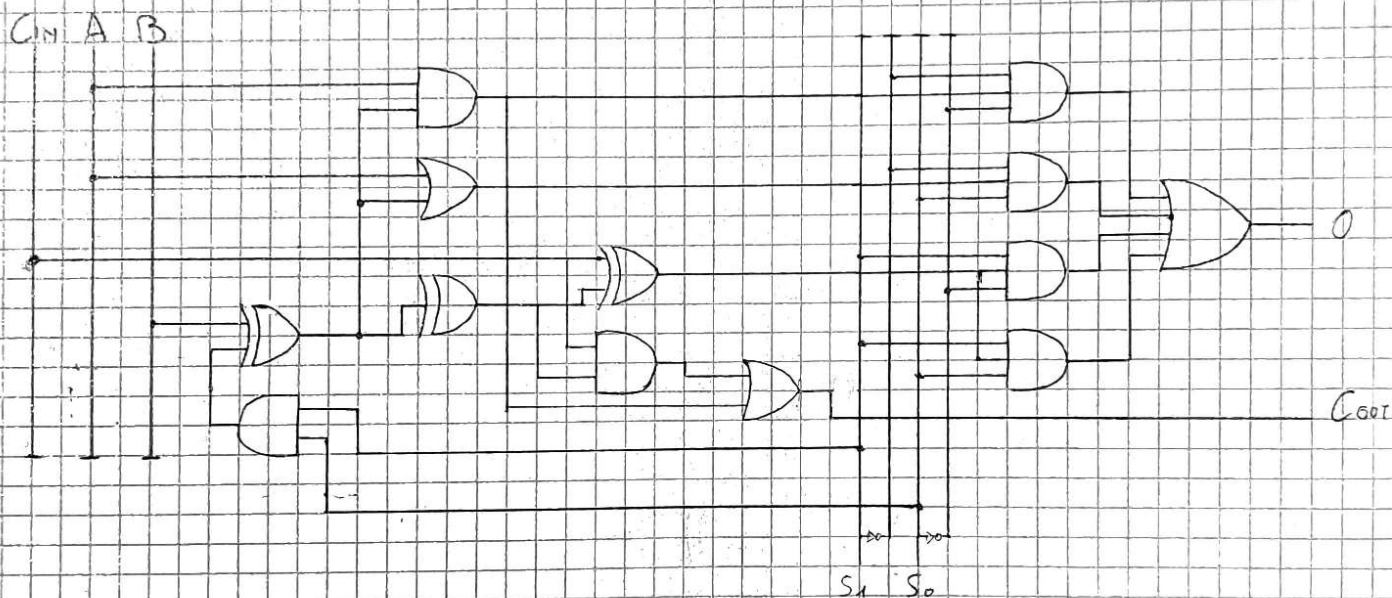
AMPLIFICATORE TRI-STATE



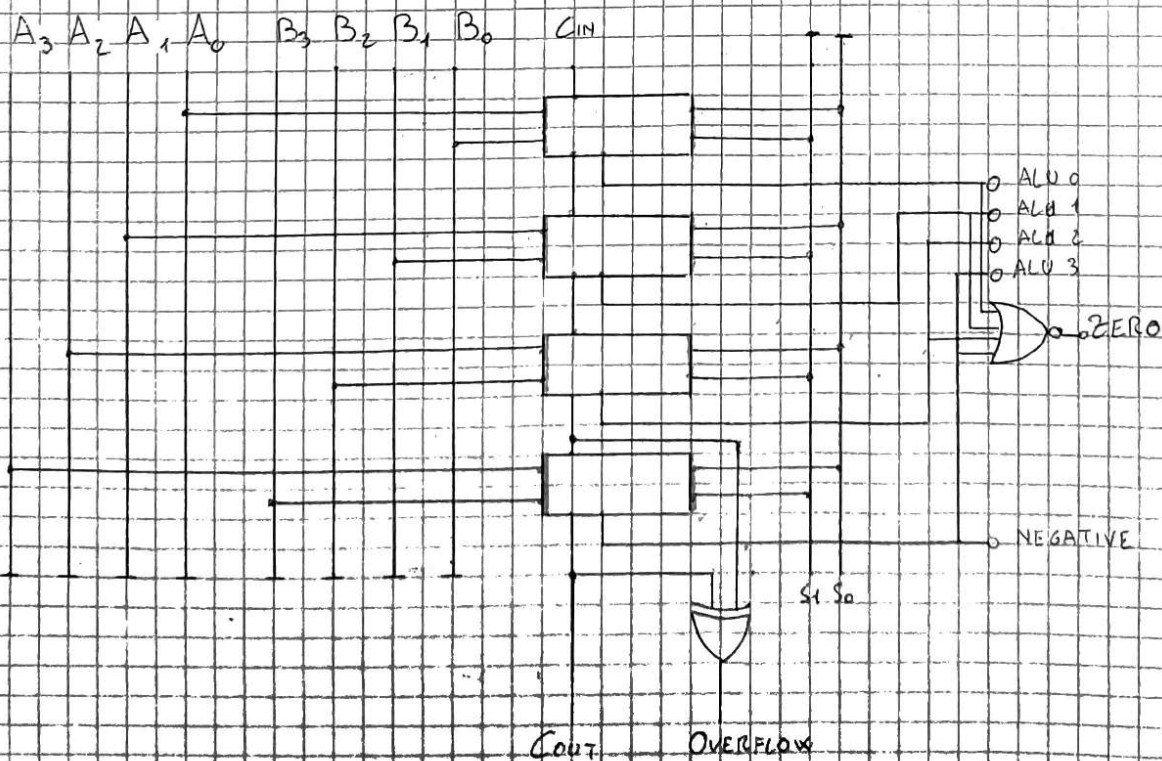
SCHEMA LOGICO ALU 1 BIT



SCHEMA LOGICO ALU 1 BIT OTTIMIZZATO



SCHEMA LOGICO ALU 4 BIT



COMPONENTI NOTEVOLI COMBINATORI

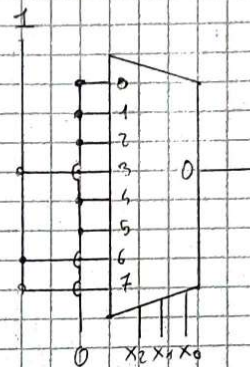
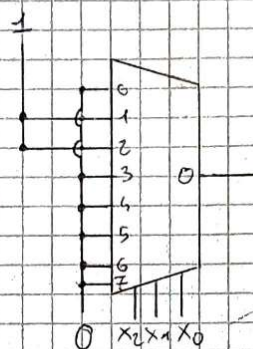
ESAME PARZIALE 4 APRILE 2023 - ES # PARTE 1

$$S_0 = \bar{X}_2 X_1 X_0 + X_2 X_1 \bar{X}_0 + X_2 X_1 X_0$$

$$S_1 = \bar{X}_2 \bar{X}_1 X_0 + \bar{X}_2 X_1 \bar{X}_0$$

IMPLEMENTAZIONE TRAMITE MULTIPLEXER 8:1:

X_2	X_1	X_0	S_1	S_0
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	0	0
1	0	1	0	0
1	1	0	0	1
1	1	1	0	1



S1

S0

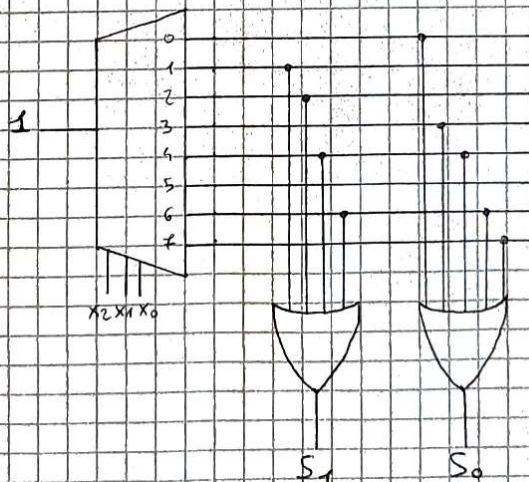
ESAME PARZIALE 4 APRILE 2023 - ES 14 PARTE 2

$$S_0 = \bar{X}_2 \bar{X}_1 \bar{X}_0 + \bar{X}_2 X_1 X_0 + X_2 \bar{X}_1 \bar{X}_0 + X_2 X_1 \bar{X}_0 + X_2 X_1 X_0$$

$$S_1 = \bar{X}_2 \bar{X}_1 X_0 + \bar{X}_2 X_1 \bar{X}_0 + X_2 \bar{X}_1 X_0 + X_2 X_1 \bar{X}_0$$

IMPLEMENTAZIONE TRAMITE DEMULTIPLEXER 1:8

X_2	X_1	X_0	S_1	S_0
0	0	0	0	1
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	1
1	0	1	0	0
1	1	0	1	1
1	1	1	0	1



• ULTERIORI ESERCIZI RIGUARDANTI:

1) MULTIPLEXER, DEMULTIPLEXER;

2) HALF ADDER, FULL ADDER;

3) ALU (ARITHMETIC LOGIC UNIT);

LABORATORIO

LOGISIM.

SONO STATI SVOLTI AL P.C. CON L'AUSILIO DEL TOOL LOGISIM.

ESAME PARZIALE 8 APRILE 2024 - ES 6 PARTE 1, ES 7 PARTE 2

$F_0 = \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}C\overline{D} + \overline{A}B\overline{C}\overline{D} + \overline{A}BC\overline{D} + \overline{A}BCD$

$F_1 = \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}CD + \overline{A}BCD + A\overline{B}\overline{C}\overline{D} + ABCD$

A	B	C	D	F ₁	F ₀
0	0	0	0	0	1
0	0	0	1	1	0
0	0	1	0	1	1
0	0	1	1	0	0
0	1	0	0	0	0
0	1	0	1	0	0
0	1	1	0	1	0
0	1	1	1	0	0
1	0	0	0	0	1
1	0	0	1	0	0
1	0	1	0	0	0
1	0	1	1	0	0
1	1	0	0	1	0
1	1	0	1	0	0
1	1	1	0	0	1
1	1	1	1	1	1

F₀ IMPLEMENTATO TRAMITE MUX 8:1

F₁ IMPLEMENTATO TRAMITE DEMUX 1:8

